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Articles

Protein subcellular localization based on deep image features and criterion learning strategy

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Abstract

The spatial distribution of proteome at subcellular levels provides clues for protein functions, thus is important to human biology and medicine. Imaging-based methods are one of the most important approaches for predicting protein subcellular location. Although deep neural networks have shown impressive performance in a number of imaging tasks, its application to protein subcellular localization has not been sufficiently explored. In this study, we developed a deep imaging-based approach to localize the proteins at subcellular levels. Based on deep image features extracted from convolutional neural networks (CNNs), both single-label and multi-label locations can be accurately predicted. Particularly, the multi-label prediction is quite a challenging task. Here we developed a criterion learning strategy to exploit the label–attribute relevancy and label–label relevancy. A criterion that was used to determine the final label set was automatically obtained during the learning procedure. We concluded an optimal CNN architecture that could give the best results. Besides, experiments show that compared with the hand-crafted features, the deep features present more accurate prediction with less features. The implementation for the proposed method is available at https://github.com/RanSuLab/ProteinSubcellularLocation.

Keywords: protein subcellular location; deep neural network; criterion learning; phenotype features

Introduction

The spatial distribution of proteins at a subcellular level reveals valuable information for understanding the working of proteins [16]. These subcellular locations provide a specific chemical environment and set of interaction partners that enable the fulfilling of the function of the proteins [29]. Abnormal subcellular locations of proteins have been reported to be associated with cellular dysfunction and disease [15, 19]. Therefore, exact protein location prediction at subcellular level can improve tasks

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such as target identification in drug discovery [13], especially for anticancer research [17], etc. Using biological experiments and visual inspection to determine the subcellular localization of a protein is reliable but suffers from large labor and time cost. Thus, computational approaches have been developed for automated subcellular localization of proteins. In terms of the data source, some computational methods predicted subcellular location using 1D protein amino acid sequence [34, 41]. Compared to the sequencing data, 2D images that present proteins or their subcellular location with distinct patterns are more intuitive and interpretable [38]. Particularly, with the advancement of imaging technologies, considerable progress has been made for the development of imaging-based methods to automatically determine the protein subcellular locations [12, 18].

Large number of hand-crafted image features have been employed to predict subcellular location of proteins [1, 14]. Texture feature, which describes the spatial arrangement of color or intensities in images, is one of the most popular image feature types that has been used in subcellular location identification. In a number of studies, such as Boland and Murphy [1], Tahir and Khan [26] and Chebira *et al.* [2], Haralick texture features, calculated based on gray level co-occurrence matrices in different orientations, were extracted to recognize the subcellular localization. Local binary pattern (LBP), which encodes the local structure around each pixel with decimal numbers, is another type of texture features that has been used extensively to predict protein subcellular localization [28, 37, 40]. Extensions of LBP such as local ternary pattern (LTP) features and local quinary pattern (LQP) features were employed in [40] to extract the texture information for location classification. Texture features such Gabor features or wavelet features were also employed in some studies to recognize protein subcellular location [9]. The 2nd commonly used type of features is the Zernike feature. Due to its invariance to the translation and rotation of images, some works such as [1, 27] extracted Zernike features to localize protein at subcellular levels. Other types of features including morphological features (the charateristics such as number intensity, shape, position,etc.) [2, 3, 8], DNA distribution features (DNA spatial distributions of a human cell) [37, 40], texton-based features [26], etc. were also widely used.

However, most of the current studies focus on localizing proteins with single labels. Each protein was assumed to correspond to only one subcellular location in these studies. In fact, at least 20% of human proteins exist at more than two subcellular locations [42]. Some studies proposed algorithms to label proteins with multiple subcellular structures. In Xu *et al.* [37], binary relevance (BR) was used to build the multi-label predictor, which treated the multi-location issue as multiple independent binary classification issues. However, the correlation among classes was not considered in this method. Xu *et al.* solved this problem by means of a Bayesian graph model and used it to guide the order of binary classifiers training and also used it as additional features [36]. Wang and Li [33] extended the BR method and learned the label correlation through feature space transformation. For each label, multiple binary classifiers were learned by randomly selecting several labels as its additional input features and then these classifiers were aggregated by majority voting strategy. Yang *et al.* [39] used frequency features and a chained prediction model to handle multi-label problem. However, the prediction performance is still waiting to be improved, as the best accuracy among all these methods is still below 70%.

Deep learning has shown impressive performance in a wide range of areas [10, 25]. Compared to the hand-crafted features that depend on prior knowledge and manual design, deep learning automatically learns features used for classification issues. As far as we know, only a handful of studies employed deep learning for protein subcellular location identification. Pärnamaa and Parts [20] trained an 11-layer neural network to classify fluorescent protein subcellular locations in yeast cells. Shao *et al.* [23] applied features extracted from the last fully connected layer of the convolutional neural networks (CNNs), selected the most distinguishing features based on Lasso model and used these selected features for final classifications. Winsnes *et al.* [35] used deep neural networks (DNNs) for predicting subcellular localization on the images of U2OS cell lines.

It is expected that the deep learning-based methods are able to achieve more progress in prediction of protein subcellular location. In this study, we proposed a protein subcellular location prediction approach that can handle both single-label and multi-label samples based on deep features extracted from CNNs. Used as feature extractors, five popular CNNs were employed and five sets of 128-dimensional feature vector were produced from them. Then we handled the five networks, respectively, and identified an optimal one. Next, we obtained a more compact feature subset through feature selection. Then the classification was made based on these optimal features. Particularly, multi-label localization is quite a challenging task. Most of the current methods lack the sufficient exploration of label relevance; thus, the accuracy is limited. To overcome these shortcomings, we proposed an approach that can learn both label–attribute relevance and label–label relevance automatically. Firstly, we generated a probability vector containing the predicted probability of each class. Then inspired by Xu *et al.* [37], we judged the label set of each sample through counting the number of 'maximal probabilities'. Different from Xu *et al.*, which depended on manual settings, the counting procedure of the proposed method was completed via automated learning of a criterion. We learned a threshold value for this criterion to determine the labels. We designed a series of experiments to validate the performance of the proposed method and the results have proved the effectiveness of the proposed method. The implementation for the proposed method is available at https://github.com/RanSuLab/ProteinSubcellularLocation.

Materials and methods

Overview of the proposed approach

The overall framework of the proposed method is shown in Figure 1. Three steps are involved: Phase A includes data collection and pre-processing; feature extraction and selection are conducted in Phase B; In Phase C, the labels of the samples are predicted. Here we handle the single-label samples and multi-label samples separately.

Phase A: data collection and pre-processing

We used the immunohistochemistry (IHC) microscopy images extracted from the Human Protein Atlas (HPA) (http://www.proteinatlas.org/) for training and validation [31]. The HPA consists of six parts: the Tissue Atlas, the Cell Atlas, the Pathology Atlas, the Blood Atlas, the Brain Atlas and the Metabolic Atlas. We extracted IHC images from the Tissue Atlas, which presents the distribution of the proteins across all major tissues and organs in the human body. Image labels were obtained from the HPA and Universal Protein Resource (UniProt; https://www.uniprot.org/) [5]. We discarded IHC images with protein expression level score 'not detected'.

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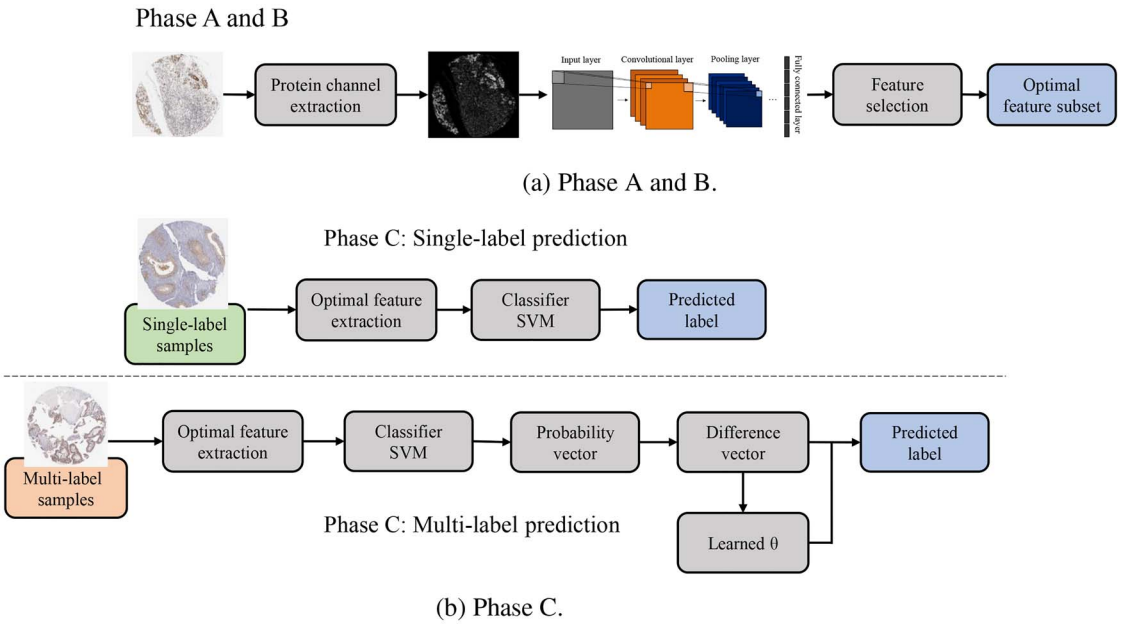


FIG. 1. Overview of the proposed method. Phase A: data collection and pre-processing; Phase B: feature extraction and selection; Phase C: subcellular location prediction. In Phase C, we deal with single-label samples (top) and multi-label samples (bottom) separately.

Single-label data and mixed-label data (including both single-label samples and multi-label samples) were both used in our study. The single-label data contains 14 antibody proteins with a total of 1386 IHC images belonging to 7 subcellular locations, which include endoplasmic reticulum (ER), cytoskeleton, Golgi apparatus, mitochondria, nucleolus, nucleus, and vesicles. In addition to these 14 antibody proteins, the mixed-label data has another 24 antibody proteins and totally contains 3129 IHC images (531 for multi-label proteins and another new 1212 single-label images besides the original 1386 images). These proteins distribute at nine subcellular locations including lysosome and cytoplasm in addition to the above seven locations. We selected the proteins with high and medium reliability scores. In order to increase the volume of training data and avoid over-fitting, we conducted data augmentation through flipping (up/down and left/right) and rotation ($\frac{\pi}{6}$, $\frac{\pi}{3}$, $\frac{\pi}{2}$, $\frac{2\pi}{3}$, $\frac{5\pi}{6}$ and π) operations. Totally, we obtained 30 910 images for the single-label data and 27 492 images for the mixed-label data. The details of the data can be found in Supplementary File 1.

The HPA data include two channels: DNA channel (purple) and protein channel (brown). We separated the protein channel from the DNA channel based on linear spectral separation (LIN) algorithm [37].

Phase B: feature extraction and selection

In our study, the deep CNNs worked as feature extractors to learn rich and discriminative information for protein subcellular location prediction. Here we extracted five sets of features from five popular networks: AlexNet [11], VggNet [24], Xception [4], ResNet [6] and DenseNet [7]. For AlexNet, VggNet and ResNet, we connected three fully connected layers with 1024, 128 and 7 nodes (9 nodes for multi-label classification); while for Xception and DenseNet, we had two fully connected layers with nodes 128 and 7 (9 nodes for multi-label classification). We stopped training when the loss decayed below a magnitude of 1%. We extracted features from the 2nd last fully connected layer and thus 128-dimensional feature vectors were generated for each architecture. Finally, each protein expression image was represented by the 128-dimensional feature vector under each CNN framework.

Then we performed feature selection to improve the computation efficiency. Here we used a minimum-redundancy maximum-relevance (mRMR) [22] algorithm combined with backward feature elimination (BFE) to reduce feature redundancy. The mRMR ranked the features based on feature mutual information and the BFE generated feature subsets by eliminating last ranked features. Then the best performed feature subset was picked as the optimal selected features.

Phase C: subcellular location prediction

For the single-label samples, the prediction is an ordinary routine. The support vector machine (SVM) was used to distinguish the subcellular locations after the optimal feature subset was determined. Different from the single-label classification, multi-label predictors must consider the relationships between labels and features and the dependencies between labels due to label co-occurrences [30]. Here we used a criterion learning strategy to find the relationship between labels and features and between labels and labels.

The iLocator proposed by Xu *et al.* [37] employed top criterion and threshold criterion to decide the final multi-label set. However, the two criterions needed to be adjusted manually to obtain an ideal result. We proposed a multi-label prediction approach, which enables automated decision of the label set without human intervention. Only one criterion was used to determine the label set and this threshold can be automatically learned.

For the multi-label prediction, each sample might be classified with one or more labels. We denote the label set for a sample as $L = \{l_1, l_2, ..., l_9\}$. If l_i belongs to this sample, the value will be set to 1. Otherwise, it is 0. We trained an SVM classifier based on the training images. Then for each image, the SVM output

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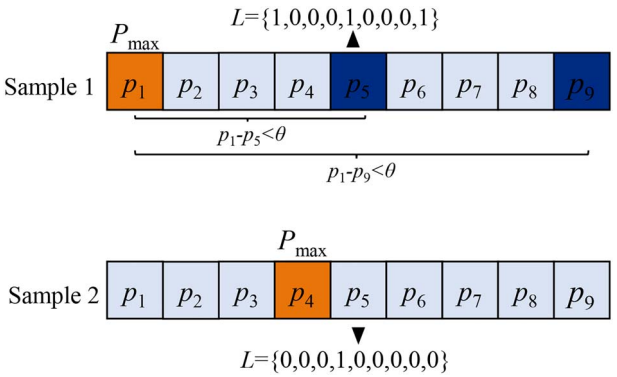


FIG. 2. The labeling strategy of the proposed method. Sample 1 is predicted to have three labels and Sample 2 is predicted to be single label. The orange boxes represent p_{max}, the dark blue boxes show the probability values similar to p_{max} and the light blue boxes represent the probability values that have large difference with p_{max}. The obtained labels L are listed in the top and bottom, respectively.

nine scores $P = \{p_1, p_2, ..., p_9\}$, which represented the probability of each class. Among them, the label with the maximum probability p_{max} should be assigned to the sample. Thus, the position in L corresponding to p_{max}'s position was set to 1. Then we sought probability values similar to p_{max}. The rationale is that true labels are supposed to have similar probabilities, which are larger than the false labels. Therefore, the ith label, which had a similar probability with p_{max}, or in other words, had a sufficiently small difference or close relevancy of probability score with p_{max}, could be assigned to the predicted image. We defined a parameter, θ, that worked as a criterion, to determine the final labels. Through comparing the value of θ with the difference of probability, the labels can be obtained. The value of l_i is determined as follows:

$$l_i = \begin{cases} 1 & \text{if } p_i = p_{max}, \\ 1 & \text{if } p_i \neq p_{max} \& p_{difi} = p_{max} - p_i < \theta, \\ 0 & \text{if } p_i \neq p_{max} \& p_{difi} = p_{max} - p_i \geqslant \theta. \end{cases} \quad (1)$$

where $P_{dif} = \{p_{dif1}, p_{dif2}, ..., p_{dif9}\}$ is the probability difference vector. If all the differences (except the p_{max} position) between p and p_{max} are larger than θ, or no elements in P_{dif} (except the p_{max} position) is smaller than θ, eight values of L are 0 and the sample is considered to be single label. We show example of labeling strategy in Figure 2.

In contrast to Xu *et al.* study, which set the two criterions manually, we only employed one criterion, θ. The value of θ was obtained through grid searching. Since the probability ranges from 0 to 1, θ was tuned from 0 to 1 with a step size of 0.05. Then the optimal θ was picked when the performance achieved the best.

The pseudo-code of the proposed method to predict multi-label samples is shown in Algorithm 1.

Evaluation metrics

We used four metrics to evaluate the proposed framework, accuracy (ACC), sensitivity (SEN), specificity (SPE) and F1-score (F1). They are defined as follows:

$$ACC = \frac{TP+TN}{TP+TN+FP+FN} \quad (2)$$

Algorithm 1 Construction of model of predicting protein's multiple subcellular locations.

Input:
1: Input: IHC images of proteins.
Output:
2: Output: Model of predicting protein subcellular locations.
3: Separate the two channels of the IHC images and use the protein channel images.
4: Augment all the images to make them sufficient in volume for training.
5: Split the images into C_{train} and C_{test}.
6: Feed C_{train} into different CNNs and test on C_{test}.
7: Extract 128-dimension feature vector F_D from the last second fully connected layer for all the images.
8: Rank the F_D based on mRMR (increasing order) and denote the rank as F_r. Make $F_{update} = F_r$.
9: Split the images into I_{train}, I_{test}, and I_{val}.
10: **for** $j = 1; j \leq 128; j++$ **do**
11: Train SVM based on I_{train}.
12: Predict subcellular locations based on F_{update} using SVM on I_{test}; Record the results as $acc[j]$, $sen[j]$, etc.
13: $F_{update} \leftarrow F_r - F_r[j]$.
14: **end for**
15: Choose the feature subset with the highest accuracy as the final selected subset, f_s.
16: Train SVM using $I_{train} + I_{test}$ based on f_s and output probability vector P; Find p_{max}.
17: Calculate difference vector P_{dif} and tune θ.
18: Test and validate the model on I_{val}.

TABLE 1. Hyper-parameter setting of the five deep networks

	LR[1]	BS[2]	BN[3]	Optimizer
AlexNet	0.01	128	Yes	Adam
VggNet	0.01	128	Yes	Adadelta
Xception	0.01	128	Yes	Adam
ResNet	0.01	64	Yes	Adadelta
DenseNet	0.01	32	Yes	Momentum

[1]LR means learning rate.[2]BS means batch size.[3]BN means batch normalization.

$$SEN = \frac{TP}{TP+FN} \quad (3)$$

$$SPE = \frac{TN}{TN+FP} \quad (4)$$

$$F1\text{-score} = \frac{2\ TP}{2\ TP+FP+FN} \quad (5)$$

Here TP, FP, TN and FN are true positive, false positive, true negative and false negative, respectively. For the multi-label classification, subset accuracy (ACC_{sub}), label accuracy (ACC_{lab}) and average label accuracy (ACC_{avelab}) were also used to provide a more stringent measurement. ACC_{sub} is the fraction of samples whose predicted label set is exactly the same as the true label set. ACC_{lab} evaluates the prediction accuracy for each label and it can be used to measure the difficulty of recognizing each label [37]. ACC_{avelab} is the average of ACC_{lab} across all the labels. In addition, we used the receive operating characteristic curves (ROCs) and

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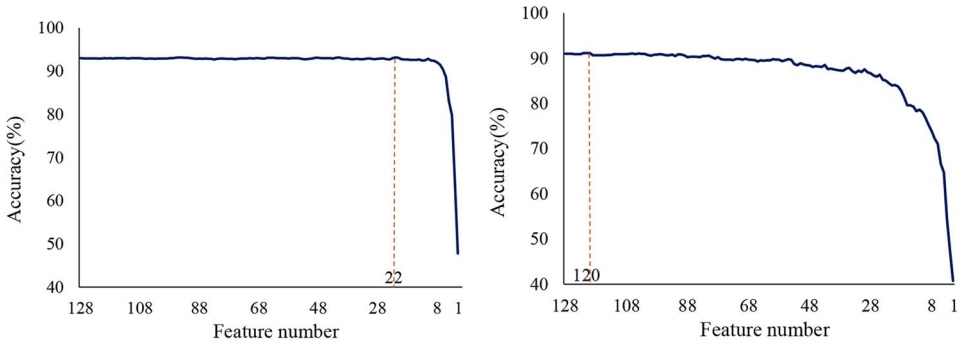


FIG. 3. The intermediate results of feature selection for single-label prediction. Here we take Xception (left) and VggNet (right) as examples. They selected 22 features and 120 features, respectively.

TABLE 2. Performance of the five deep networks on single-label samples

	FN[1]	ACC (%)	SEN (%)	SPE (%)	AUC	F1 (%)
AlexNet	95	91.4	91.2	98.5	0.99	91.7
VggNet	120	89.9	89.3	98.3	0.99	89.7
Xception	22	92.1	91.9	98.7	0.99	91.9
ResNet	108	75.7	75.3	95.9	0.96	75.1
DenseNet	51	89.2	89.6	98.2	0.99	89.8

[1]FN means feature number.

the area under the ROC (AUC) to evaluate the performance. We used 10-fold cross validation to validate the model.

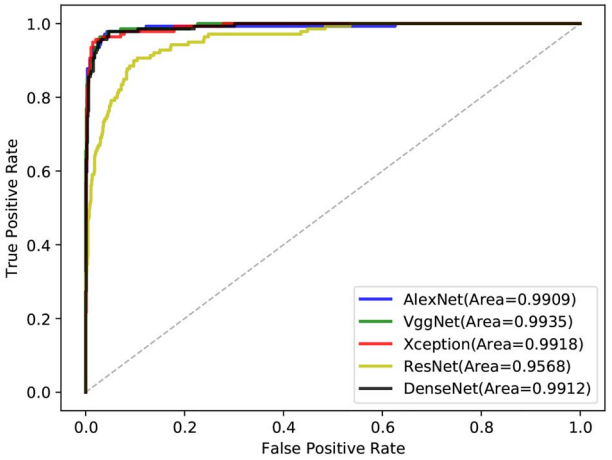


FIG. 4. The ROC curves of the five deep networks to classify single-label samples.

Results

Prediction of single-label locations

We firstly tested the performance of the proposed method using the single-label proteins. We trained the AlexNet, VggNet, Xception, ResNet and DenseNet and extracted 128-dimensional deep features from them, respectively. A total of 24 750 images were used for training the CNN, and 6160 images were used for testing. The networks were initialized randomly and the setting of the networks is shown in Table 1. Then we conducted feature selection based on mRMR/BFE and then carried out the classifications using SVM. The intermediate results of feature selection are shown in Figure 3. From the figure, it shows that similar or better performance can be obtained using smaller number of features through feature selection.

After picking the optimal feature subsets, the model was trained based on the optimal features. We show the performance of each network on the test data in Table 2. From the table, it shows that the Xception achieves the best performance. With only 22 deep features, it can reach ACC 92.1% and AUC 0.9918. Following Xception, AlexNet, VggNet, DenseNet and ResNet present a decreasing order for ACC values. We also show the ROC curves in Figure 4. It can be seen that the AlexNet, VggNet, Xception and DenseNet have similar AUC values, while ResNet stays under the other four networks. We show the confusion matrix of the best performed network, Xception in Table 3. It shows that the prediction of mitochondrion achieves 100% while predicting vesicles is not as accurate as other subcellular locations. The confusion matrices of AlexNet, VggNet, DenseNet and ResNet can be found in Supplementary File 2.

Performance of the proposed predictor to classify mixed-label locations

Discrimination of multi-label and single-label samples

The proposed approach must be able to discriminate single-label and multi-label proteins if mixed-label samples are provided. In our design, whether a sample is single-label or multi-label is judged based on the number of elements with probability difference smaller than θ in the probability difference vector P_{dif}. If the number is 0 (except p_{max} position), it means other probability scores except p_{max} are far from p_{max}; thus, this sample is single-labeled; if the number is larger than 0, it means at least one probability score is similar to p_{max} besides p_{max} itself; thus, this is a multi-label sample. Assuming the multi-label is labeled as 1 and single-label is 0, we looked into the performance of the proposed method on distinguishing single-label and multi-label samples based on mixed-label data set. The results are shown in Table 4.

From the table, it shows that VggNet reaches the highest accuracy, 93.0%, followed by AlexNet, ResNet, DenseNet and Xception. The result is different from the single-label classification as Xception performs the best in single-label prediction.

Prediction of mixed-label locations

The proposed method enables discrimination of single-label and multi-label images. We tested the performance of the proposed method for predicting the mixed-label samples. A total of 21 984 images were used to train the CNN and 5508 images for testing.

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Table 3. Confusion matrix of Xception

	Golgi apparatus (%)	Mitochondrion (%)	Vesicles (%)	ER (%)	Nucleolus (%)	Nucleus (%)	Cytoskeleton (%)
Golgi apparatus	92.9	0	0	0	7.1	0	0
Mitochondrion	0	100	0	0	0	0	0
Vesicles	0	0	81.8	4.5	0	4.5	9.1
ER	0	0	4.3	95.7	0	0	0
Nucleolus	0	0	0	0	90.9	9.1	0
Nucleus	0	0	6.7	0	6.7	86.7	0
Cytoskeleton	0	4.8	0	0	0	0	95.2

Table 4. Performance of the proposed method distinguishing single-label and multi-label samples

	FN[1]	Total N[2]	$N\left(n_{p_{difi}<\theta}=0\right)^3$	$N\left(n_{p_{difi}<\theta}\geqslant 1\right)^4$	ACC (%)
AlexNet	116	315	243	43	90.8
VggNet	116	315	244	49	93.0
Xception	22	315	186	16	64.1
ResNet	107	315	219	32	79.7
DenseNet	13	315	219	27	78.1

[1] FN means feature number.
[2] Total N means the total number of images.
[3] $n_{p_{difi}<\theta}=0$ means the difference vector that has 0 element with $p_{dif}<\theta$ besides p_{max} position and $N\left(n_{p_{difi}<\theta}=0\right)$ shows the correctly predicted number of images with the difference vector that has 0 elements with $p_{dif}<\theta$ (besides p_{max} position). In other words, this is the correctly detected number of single-label samples. Therefore, it is the true negative result.
[4] $n_{p_{difi}<\theta}\geqslant 1$ means the difference vector that has more than one element with $p_{dif}<\theta$ besides p_{max} position and $N\left(n_{p_{difi}<\theta}\geqslant 1\right)$ shows the correctly predicted number of images with the difference vector that has more than one element with $p_{dif}<\theta$ (besides p_{max} position). In other words, this is the correctly detected number of multi-label samples. Therefore, it is the true positive result.

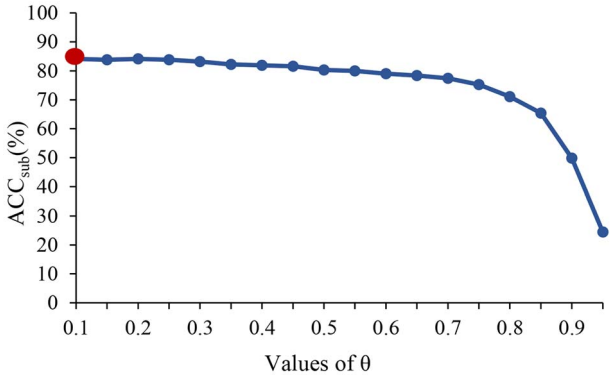


Fig. 5. The learning of θ. The starting value of θ is 0.1 and the step size is 0.05. The curves achieve peaks when θ equals to 0.1.

The mixed-label classification used the selected features and the probability vector to learn θ and this θ determines the final label set. The setting of θ was obtained through learning. Here the starting value of θ was 0.1. Then θ was incremented with a step-size 0.05 until reaching 1. The performance curves of learning θ can be found in Figure 5. We obtained the value of θ when the curves reached peaks.

Next, we tested the performance of the five networks, which are shown in Table 5 and Figure 6 (for the ROC, we obtained one graph for each class. Then we present the micro-average of the graphs of all the classes). In Table 5, it can be seen that VggNet achieves the highest subset accuracy, accuracy and AUC. However, Xception has the worst performance with the lowest subset accuracy, accuracy and AUC values. In Figure 6, VggNet stays top leftmost while Xception stays below other curves. We further show the performance of recognizing each subcellular location in Figure 7, in order to observe which location is easier to detect. It shows that 'Golgi apparatus', 'Cytoskeleton' and 'Nucleolus' stay quite close and above other locations, which means these three subcellular locations are easier to be recognized; while 'Mitochondrion' and 'Cytoplasm' are more difficult to be identified among the nine locations.

Table 5. Performance of the five deep networks on mixed-label data

	FN[1]	θ	ACC_{sub}(%)	ACC_{avelab}(%)	ACC (%)	AUC	F1 (%)
AlexNet	116	0.15	80.3	95.3	83.0	0.95	83.1
VggNet	116	0.1	84.1	95.2	85.9	0.97	82.7
Xception	22	0.1	29.8	80.6	39.5	0.77	40.6
ResNet	107	0.1	67.3	92.6	74.1	0.95	73.5
DenseNet	13	0.1	57.1	89.8	65.5	0.92	69.1

[1]FN means feature number.

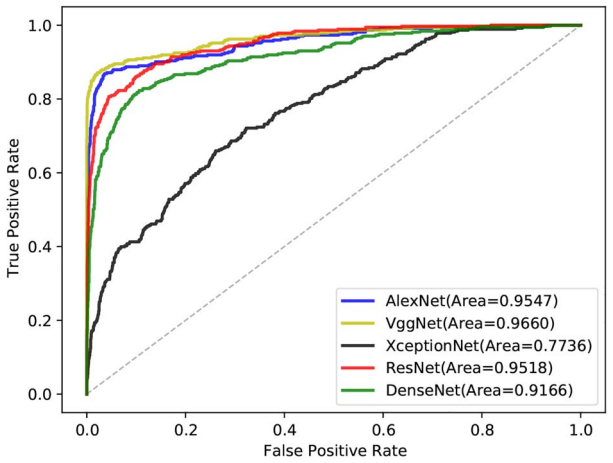


Fig. 6. The ROC curves of the five deep networks to classify mixed-label samples.

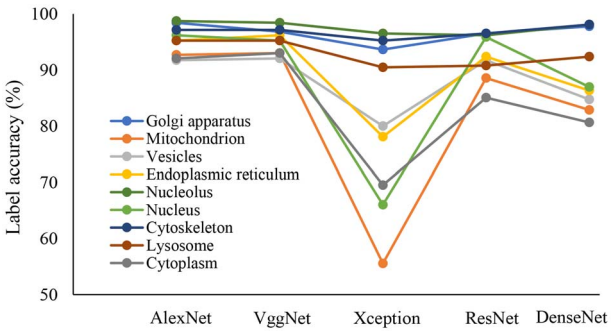
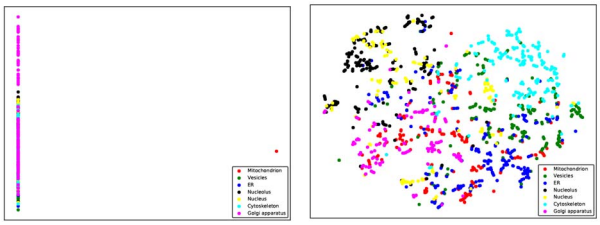


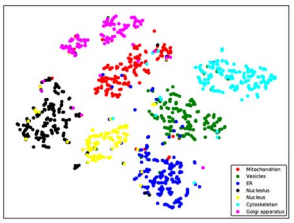
Fig. 7. The performance of recognizing each subcellular location across the five deep architectures.

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(a) t-SNE plotting of Yang et al.'s features.

(b) t-SNE plotting of VggNet.



(c) t-SNE plotting of Xception.

FIG. 8. t-SNE plotting of the deep features and hand-crafted features. The t-SNE was applied on single-label data.

TABLE 6. Comparison between the deep features and Yang *et al.*'s features for single-label samples. We list the best and worst performed CNNs, the Xception and ResNet, for comparison

	FN[1]	ACC (%)	SEN (%)	SPE (%)	AUC	F1 (%)
Xception	22	92.1	91.9	98.7	0.99	91.9
ResNet	108	75.7	75.3	95.9	0.96	75.1
Yang *et al.*'s	500	71.2	70.8	95.2	0.93	70.7

[1]FN means feature number.

Comparison with hand-crafted features

Most of the current protein subcellular localization approaches extracted hand-crafted features and conducted prediction based on these features. We extracted four types of popular texture features, which were also used in Yang *et al.*'s study as comparison studies. These features include 8126 Haralick texture features, 28 LBP features, 72 LTP features and 288 LQP features. Totally 8514 hand-crafted features were used for comparison. Similar to the procedure in the proposed method, we selected the optimal feature subset and made prediction. The details of the four types of features can be found in Supplementary File 3.

Visualization and representation of features

In order to get insights into the features, we applied the t-distributed stochastic neighbor embedding (t-SNE) [32] to visualize the distribution of deep features and hand-crafted features (denoted as Yang *et al.*'s method [40]). The plotting can be found in Figure 8. From the figure, it shows that the Xception separates the features quite well in low dimension space while VggNet becomes more mixing. Yang *et al.*'s features almost align one line and it is difficult to distinguish the seven categories even in low dimension. This has shown that the deep features are more discriminative than the hand-crafted features.

Comparison with hand-crafted features on single-label samples

We compared the performances between deep features and Yang *et al.*'s features on single-label samples. We extracted features, selected the optimal feature subset based on mRMR/BFE and predicted the labels using SVM. The results are shown in Table 6 and Figure 9.

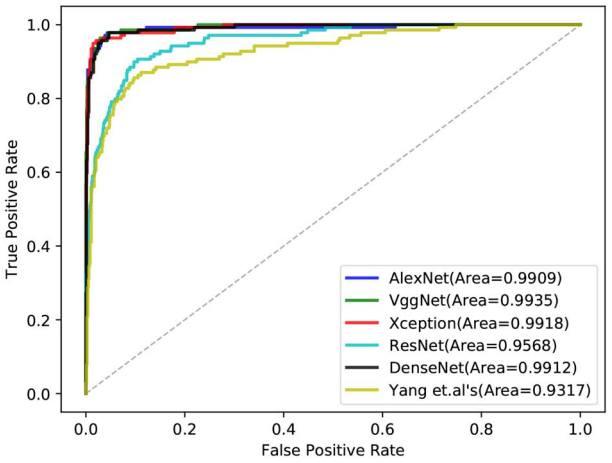


FIG. 9. The comparison between the deep features and Yang *et al.*'s features to classify single-label samples.

TABLE 7. Comparison between the deep features and Yang *et al.*'s features on mixed-label samples. We list the best and worst performed CNNs, the VggNet and Xception, for comparison

	FN[1]	ACC (%)	SEN (%)	SPE (%)	AUC	F1 (%)
VggNet	116	85.9	88.6	96.2	0.97	82.7
Xception	22	39.5	45.9	84.4	0.77	40.6
Yang *et al.*'s	480	54.9	59.2	90.9	0.91	57.4

[1]FN means feature number.

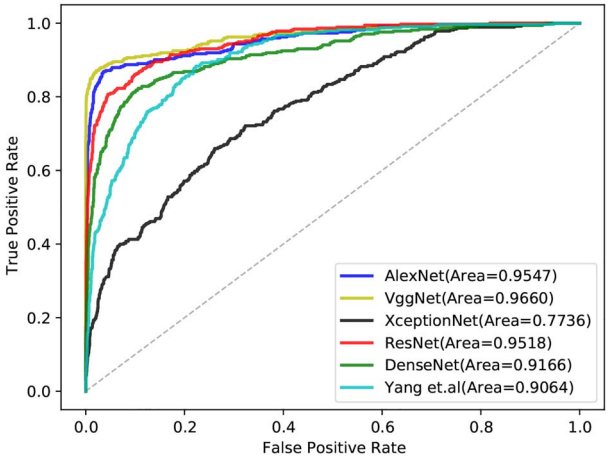


FIG. 10. The comparison between the deep features and Yang *et al.*'s features to classify mixed-label samples.

From Table 6, it can be found that the Yang *et al.*'s approach reached ACC 71.2%, which is 20% lower than the Xception and even lower than the worst performed CNN, the ResNet. Yang *et al.*'s accuracy was achieved by selected 500 features. The size of the selected feature subset is much larger than that of CNNs.

Comparison with hand-crafted features on mixed-label samples

We also compared between the proposed method with the hand-crafted features for mixed-label subcellular location prediction. The results are shown in Table 7 and Figure 7.

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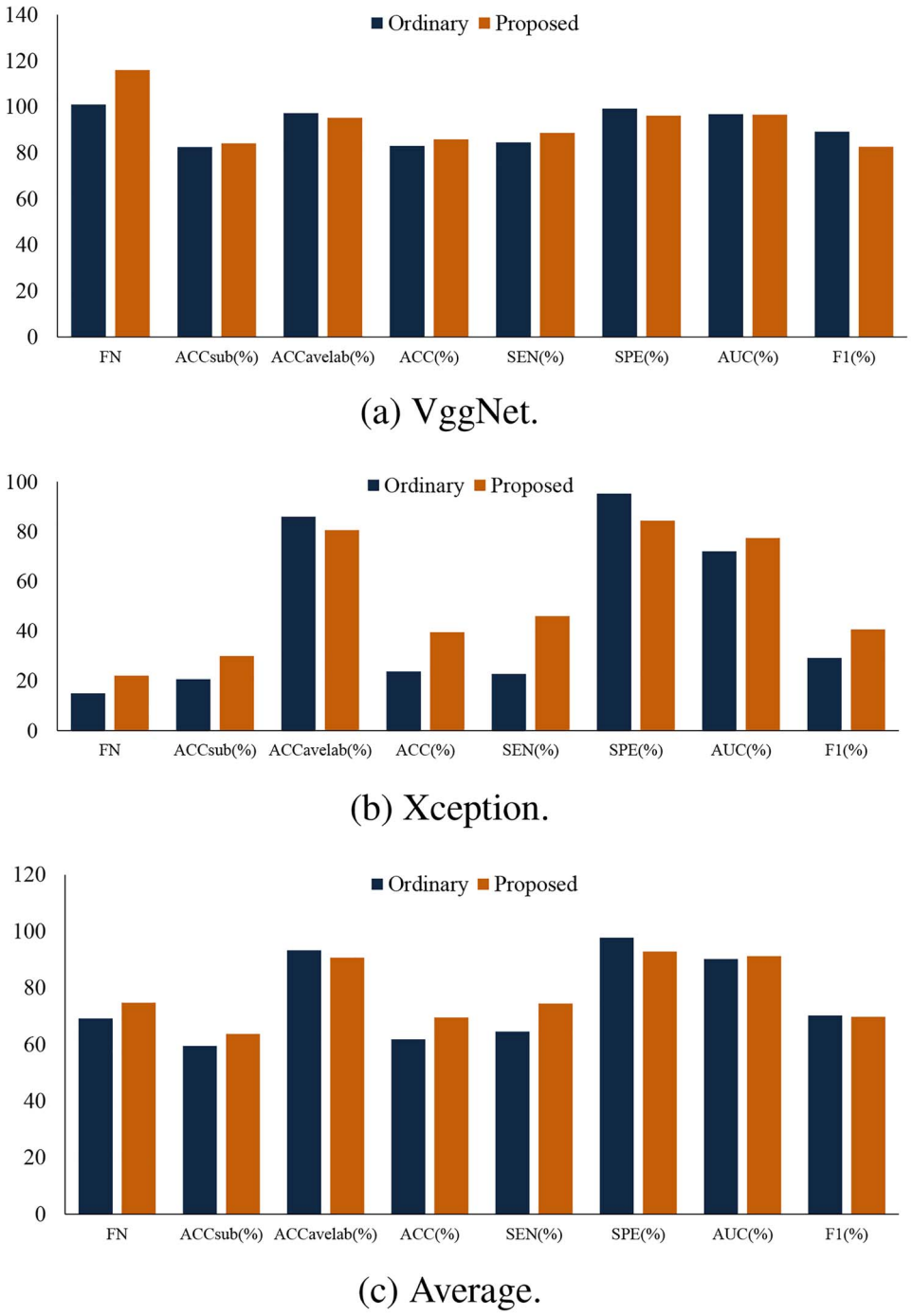


FIG. 11. The comparison between the ordinary cut-off method and the proposed method on mixed-label samples. FN represents feature number. We here take VggNet and Xception as examples. (c) shows the average comparison across the five deep networks. Each column represents the mean value of the five architectures. Comparisons of AlexNet, ResNet and DenseNet can be found in Supplementary File 4.

From Table 7 and Figure 7, it shows that Yang *et al.*'s method performed between the best and worst deep features in terms of accuracy. It has the largest feature number among the three methods. This means that for mixed-label samples, the CNN also achieves more competitive results than the hand-crafted features.

Multi-label learning strategy

In the multi-label prediction, the SVM output a probability vector. We considered the label with the maximal probability as the 1st predicted label and calculated the difference vector. Then we learned a criterion θ. Through comparing the difference

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Table 8. Performance of the five deep networks on mixed-label data using the ordinary cut-off method

	FN[1]	ACC_{sub}(%)	ACC_{avelab}(%)	ACC (%)	AUC	F1 (%)
AlexNet	109	79.4	96.3	79.9	0.96	85.1
VggNet	101	82.5	97.2	83.0	0.97	89.3
Xception	15	20.6	85.9	23.7	0.72	29.2
ResNet	108	61.6	94.5	65.6	0.95	77.9
DenseNet	13	53.3	92.6	57.0	0.91	70.0

[1]FN means feature number.

vector with the criterion, the other labels could be determined. We compared the proposed strategy with the ordinary cut-off method, that is, if the probability was larger than 0.5, the label was set to 1; otherwise, it was set to 0. The results are shown in Table. 8. We show the comparison between the ordinary method and the proposed method in Figure 11. Generally speaking, the proposed multi-label learning outperforms the ordinary cut-off for most of the metrics, which shows that the learning of proper criterion is more accurate than the pre-set threshold values.

The CNNs were implemented under Tensorflow framework (v1.14.0) in Python environment (Python foundation version 3.5.2). The classification analysis was performed using the python (v3.6) based scikit-learn (v0.19.1) [21]. The feature selection was performed using the pymrmr(v0.1.8). The hand-craft feature extraction was performed using matlab (v2018a).

Conclusions and discussions

In this study, we proposed an imaging-based method to predict protein subcellular location from IHC images. The CNNs, as one of the most popular deep learning-based approaches used in imaging tasks, worked as feature extractor to automatically learn the most informative features for prediction. In the past, only a handful of studies have tried the deep features for protein localization prediction. This study has shown the deep features are more discriminative than the traditionally used hand-crafted features.

The proposed method can handle proteins locating at both single and multiple locations. We exploited label–label relevance, which is ignored in many multi-label classification methods, and made more accurate predictions. Different from previous studies, which set two criterions manually to decide the final label set after probability vector was generated, we used a learning strategy to automatically find a proper criterion value in order to avoid human intervention.

In this study, we have tested five popular CNNs. For single-label samples, the Xception achieved the best performance while for mixed-label samples, the VggNet reached the highest accuracy. Compared with the hand-crafted features, the CNNs also show more competitive results. However, the proposed method has some limitations. Firstly, we tested only five architectures of CNNs. The results might be further improved if more architectures were employed. Secondly, the protein channel was used in the proposed method. Although most of the current studies have proved that protein channel is more effective, it is more reasonable to test both DNA channel or combination of both channels. In the future, we target to try more architectures and more channels and further improve the results. Besides, we will try to combine diverse sources of information such as the evolutionary information and physicochemical information to develop a multi-modality model to handle this problem.

Key Points

- We proposed a protein subcellular location prediction method based on phenotype features. Immunohistochemistry images were used to predict the subcellular locations of proteins.
- Although deep learning has shown impressive performance in imaging tasks, only a handful of studies have applied it to protein subcellular localization issues. We provided a conclusion that which architecture could give optimal results.
- Single-label and multi-label proteins can be treated. We here for the 1st time developed a multi-label prediction method using a criterion learning strategy. This has enabled the learning of label-label relevancy in mixed-label classification.

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Conflict of interest

The authors declared that they have no conflicts of interest to this work.

References

1. Boland MV, Murphy RF. A neural network classifier capable of recognizing the patterns of all major subcellular structures in fluorescence microscope images of HeLa cells. *Bioinformatics* 2001; **17**(12): 1213–23.
2. Chebira A, Barbotin Y, Jackson C, *et al*. A multiresolution approach to automated classification of protein subcellular location images. *BMC Bioinform* 2007; **8**:210.
3. Chen S-C, Zhao T, Gordon GJ, *et al*. Automated image analysis of protein localization in budding yeast. *Bioinformatics* 2007; **23**(13): i66–71.
4. Chollet F. Xception: deep learning with depthwise separable convolutions. In: *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA*, New York, NY, USA: IEEE, 2017, 1800–1807.
5. UniProt Consortium. UniProt: a worldwide hub of protein knowledge. *Nucleic Acids Res* 2019; **47**(D1): D506–15.
6. He K, Zhang X, Ren S, *et al*. Deep residual learning for image recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, New York, NY, USA: IEEE, 2016, 770–8.
7. Gao H, Liu Z, Van Der Maaten L, *et al*. Densely connected convolutional networks. In: *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA*, New York, NY, USA: IEEE, 2017, 2261–2269.
8. Huang K, Murphy RF. Automated classification of subcellular patterns in multicell images without segmentation into single cells. In: *Proceedings of the 2004 IEEE International Symposium on Biomedical Imaging*, New York, NY, USA: IEEE, 2004, 1139–42.

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9. Huang K, Murphy RF. Boosting accuracy of automated classification of fluorescence microscope images for location proteomics. *BMC Bioinform* 2004; **5**:78.
10. Jin Q, Meng Z, Pham TD, *et al*. DUNet: a deformable network for retinal vessel segmentation. *Knowl Based Syst* 2019;(178): 149–62.
11. Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. In: *Advances in Neural Information Processing Systems*, New York, NY, USA: Curran Associates Inc, 2012, 1097–105.
12. Lin C-C, Tsai Y-S, Lin Y-S, *et al*. Boosting multiclass learning with repeating codes and weak detectors for protein subcellular localization. *Bioinformatics* 2007; **23**(24): 3374–81.
13. Lomenick B, Olsen RW, Huang J. Identification of direct protein targets of small molecules. *ACS Chem Biol* 2011; **6**(1): 34–6.
14. Alex XL, Yolanda T, Chong ISH, *et al*. Integrating images from multiple microscopy screens reveals diverse patterns of change in the subcellular localization of proteins. *Elife* 2018; **7**:e31872.
15. Hung M-C, Link W. Protein localization in disease and therapy. *J Cell Sci* 2011; **124**(20): 3381–92.
16. Murphy RF, Boland M, Velliste M. Towards a systematics for protein subcellular location: quantitative description of protein localization patterns and automated analysis of fluorescence microscope images. In: *Proceedings of the Eighth International Conference on Intelligent Systems for Molecular Biology, La Jolla/San Diego, CA, USA*, Vol. 8. Palo Alto, CA, USA: AAAI Press, 2000, 251–259.
17. Murphy RF. Automated interpretation of protein subcellular location patterns: implications for early cancer detection and assessment. *Ann N Y Acad Sci* 2004; **1020**(1): 124–31.
18. Murphy RF, Velliste M, Porreca G. Robust numerical features for description and classification of subcellular location patterns in fluorescence microscope images. *J VLSI Signal Process Syst Signal Image Video Technol* 2003; **35**:311–21.
19. Park S, Yang J-S, Shin Y-E, *et al*. Protein localization as a principal feature of the etiology and comorbidity of genetic diseases. *Mol Syst Biol* 2011; **7**:494.
20. Pärnamaa T, Parts L. Accurate classification of protein subcellular localization from high-throughput microscopy images using deep learning. *G3 (Bethesda)* 2017; **7**(5): 1385–92.
21. Pedregosa F, Varoquaux G, Gramfort A, *et al*. Scikit-learn: machine learning in python. *J Mach Learn Res* 2011; **12**: 2825–30.
22. Peng H, Long F, Ding C. Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy. *IEEE Trans Pattern Anal Mach Intell* 2005; **27**(8): 1226–38.
23. Shao W, Ding Y, Shen H-B, *et al*. Deep model-based feature extraction for predicting protein subcellular localizations from bio-images. *Front Comput Sci* 2017; **11**(2): 243–52.
24. Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. In: *Proceedings of International Conference on Learning Representations*, 2015.
25. Ran S, Wu H, Liu X, *et al*. Predicting drug-induced hepatotoxicity based on biological feature maps and diverse classification strategies. *Brief Bioinform* 2020.
26. Tahir M, Khan A. Protein subcellular localization of fluorescence microscopy images: employing new statistical and Texton based image features and SVM based ensemble classificatio. *Inform Sci* 2016; **345**:65–80.
27. Tahir M, Khan A, Majid A. Protein subcellular localization of fluorescence imagery using spatial and transform domain features. *Bioinformatics* 2012; **28**(1): 91–7.
28. Tahir M, Khana A, Majid A, *et al*. Subcellular localization using fluorescence imagery: utilizing ensemble classification with diverse feature extraction strategies and data balancing. *Appl Soft Comput* 2013; **13**(11): 4231–43.
29. Thul PJ, Akesson L, Wiking M, *et al*. A subcellular map of the human proteome. *Science* 2017; **356**(6340): eaal3321.
30. Tsai C-P, Lee H-Y. Adversarial learning of label dependency: a novel framework for multi-class classification. In: *Proceedings of 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Brighton, United Kingdom*, New York, NY, USA: IEEE, 2019, 3847–3851.
31. Uhlen M, Cheng Z, Sjöstedt E, *et al*. A pathology atlas of the human cancer transcriptome. *Science* 2017; **357**(6352): eaan2507.
32. van der Maaten LJP, Hinton GE. Visualizing high-dimensional data using t-SNE. *J Mach Learn Res* 2008; **9**: 2579–605.
33. Wang X, Li G-Z. Multilabel learning via random label selection for protein subcellular multilocations prediction. *IEEE/ACM Trans Comput Biol Bioinform* 2013; **10**(2): 436–46.
34. Wei L, Ding Y, Ran S, *et al*. Prediction of human protein subcellular localization using deep learning. *J Parallel Distrib Comput* 2018; **117**:212–7.
35. Winsnes C, Sullivan DP, Smith K, *et al*. Multi-label prediction of subcellular localization in confocal images using deep neural networks. *Mol Biol Cell* Rockville, MD, USA: American Society for Cell Biology, 2016; **27**.
36. Xu Y-Y, Yang F, Shen H-B. Incorporating organelle correlations into semi-supervised learning for protein subcellular localization prediction. *Bioinformatics* 2016; **32**(14): 2184–92.
37. Xu Y-Y, Yang F, Zhang Y, *et al*. An image-based multi-label human protein subcellular localization predictor (iLocator) reveals protein mislocalizations in cancer tissues. *Bioinformatics* 2013; **29**(16): 2032–40.
38. Xu Y-Y, Yao L-X, Shen H-B. Bioimage-based protein subcellular location prediction: a comprehensive review. *Front Comput Sci* 2018; **12**:26–39.
39. Yang F, Yang L, Wang Y, *et al*. $\text{mic}_{\text{locator}}$: a novel image-based protein subcellular location multi-label prediction model based on multi-scale monogenic signal representation and intensity encoding strategy. *BMC Bioinform* 2019; **20**:522.
40. Yang F, Xu Y-Y, Wang S-T, *et al*. Image-based classification of protein subcellular location patterns in human reproductive tissue by ensemble learning global and local features. *Neurocomputing* 2014; **131**:113–23.
41. Zhou H, Yang Y, Shen H-B. Hum-mPLoc 3.0: prediction enhancement of human protein subcellular localization through modeling the hidden correlations of gene ontology and functional domain features. *Bioinformatics* 2017; **33**(6): 843–53.
42. Lin Z, Yang J, Shen H-B. Multi label learning for prediction of human protein subcellular localizations. *Protein J* 2009; **28**(9–10): 384–90.

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